

ECE 635 HWK4 Answers

1.(a) $\lambda = 0.628 \text{ m}$; $f = 4.78 \times 10^8 \text{ Hz}$, $\omega = 3 \times 10^9 \text{ rad/s}$

(b) $\vec{E}_z^i = 10 \cos(3 \times 10^9 t - 6x - 8z) \hat{y} \text{ V/m}$

$\vec{H}^i = (-0.021 \hat{x} + 0.016 \hat{z}) \cos(3 \times 10^9 t - 6x - 8z) \text{ A/m}$

To get time domain $\vec{H}^i$, express $\vec{E}^i$ in the frequency domain & use ME's to get $\vec{H}^i$. Then convert $\vec{H}^i$ to $\vec{H}^i$.

(c) $\theta_i = 36.9^\circ$

(d) $\vec{E}^{\text{tot}} = \vec{E}^i + \vec{E}^r = 10e^{-j(6x+8z)} \hat{y} - 10e^{-j(6x-8z)} \hat{y}$

$= -j20e^{-j6x} \sin(8z) \hat{y}$

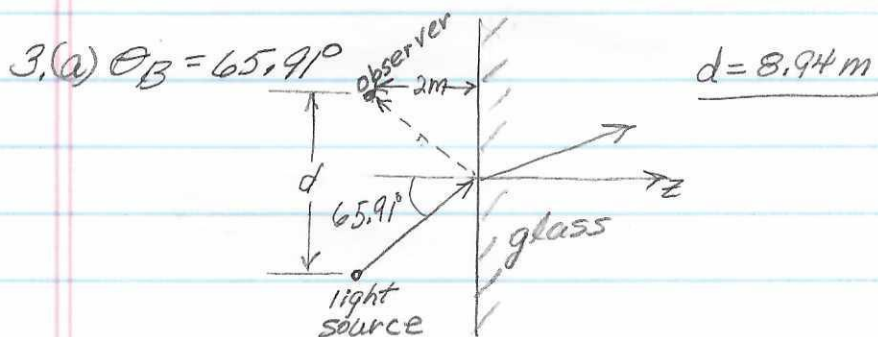
$\vec{H}^{\text{tot}} = -\left[\frac{1}{7.5\pi} e^{-j6x} \cos(8z) \hat{x} - \frac{j}{10\pi} e^{-j6x} \sin(8z) \hat{z} \right]$

2.(a) $\theta_c = 51.67^\circ \Rightarrow \theta_i$ confined to $51.67^\circ \leq \theta \leq 90^\circ$

(b) minimum angle is $\theta_i = \theta_c = 51.67^\circ$

(c) at 65° $\alpha_{2c} = 4.59 \times 10^{+6}$

$3dp = 3/\alpha_{2c} = 6.52 \times 10^{-7} \text{ m}$



(b) parallel polarization

(c) $n_2 = \sqrt{\epsilon_r} = \sqrt{5} = 2.236$